

Co⁺ ionisation limit 63564.60 cm⁻¹ = 7.88101 eV (157.320 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $\langle 0 || er || j \rangle$ in a_0
 Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model or semi-empirical calculation.
 Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
 Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ \langle J U J\rangle $ (e.a.)	A (s ⁻¹)	$\Gamma/2\pi$ ur. (MHz)	branch (%)	Use
$a^1F_{3/2} - w^1F_{3/2}^o$	233.9379	233.8662	1281.5042	1.525	9.200e+07	36.17	40.5	
$a^1F_{5/2} - x^1G_{5/2}^o$	240.7983	240.7251	1244.9940	6.391	4.940e+08	75.58	98.9	
$a^1F_{5/2} - x^1G_{3/2}^o$	241.5191	241.4457	1241.2787	4.856	4.240e+08	75.79	98.8	
$a^1F_{3/2} - x^1G_{3/2}^o$	241.6024	241.5290	1240.8506	3.999	3.830e+08	69.2	82.8	
$a^1F_{3/2} - x^1F_{5/2}^o$	242.5668	242.4931	1235.9172	5.112	3.710e+08	63.66	96.1	
$a^1F_{3/2} - x^1F_{3/2}^o$	243.2947	243.2209	1232.1932	4.259	3.190e+08	58.95	86.1	
$a^1F_{3/2} - x^1F_{5/2}^o$	243.7396	243.6657	1232.9702	3.313	2.560e+08	56.84	71.7	
$a^1F_{3/2} - x^1F_{3/2}^o$	243.9778	243.9038	1228.7695	2.838	2.810e+08	54.88	81.5	
$b^1F_{3/2} - w^1F_{3/2}^o$	251.1774	251.1018	1193.5487	2.653	9.000e+07	28.42	50.4	
$a^1F_{3/2} - x^1D_{3/2}^o$	252.2122	252.1363	1188.6519	4.242	2.840e+08	51.34	88	
$a^1F_{3/2} - x^1D_{5/2}^o$	252.9727	252.8967	1185.0781	3.537	2.610e+08	49.74	83.5	
$a^1F_{3/2} - x^1D_{3/2}^o$	253.6724	253.5961	1181.8097	2.723	2.300e+08	46.81	78.2	
$a^1F_{3/2} - x^1D_{5/2}^o$	254.5017	254.4253	1177.9586	2.150	2.840e+08	46.81	76.6	
$b^1F_{3/2} - y^1F_{3/2}^o$	340.6091	340.5114	880.1658	4.674	1.120e+08	23.07	97.3	
$b^1F_{3/2} - y^1G_{5/2}^o$	345.4498	345.3508	867.8323	5.677	1.320e+08	21.51	97.7	
$b^1F_{3/2} - y^1D_{5/2}^o$	350.7315	350.6312	854.7635	3.484	9.500e+07	18.29	82.7	

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
 $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

⁵⁹Co hyperfine structure ($I = 7/2$)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (mHz)							
a ¹ F _{5/2}	450.2835(9)	139.22(3)	F=1: -8604.15	F=2: -7723.47	F=3: -6398.31	F=4: -4623.69	F=5: -2392.99	F=6: +302.08	F=7: +3471.47	F=8: +7126.77
a ³ F _{7/2}	490.5672(31)	94.50(3)	F=0: -7675.82	F=1: -7194.90	F=2: -6231.12	F=3: -4780.62	F=4: -2837.64	F=5: -394.44	F=6: +2558.61	F=7: +4603.08
a ³ F _{5/2}	613.348(3)	67.54(6)	F=1: -6863.99	F=2: -5956.59	F=3: -3838.25	F=4: -1400.30	F=5: +1668.86	F=6: +5383.68		
a ³ F _{3/2}	1042.9828(9)	67.618(18)	F=2: -7003.91	F=3: -3923.26	F=4: +229.35	F=5: +5492.56				
b ¹ F _{5/2}	828.800(3)	-118.7(3)	F=1: -16008.82	F=2: -14334.26	F=3: -11825.95	F=4: -8488.14	F=5: -4326.47	F=6: +651.99	F=7: +6438.75	F=8: +13023.92
b ³ F _{7/2}	668.918(3)	-79.23(21)	F=0: -10577.91	F=1: -9900.90	F=2: -8548.51	F=3: -6523.97	F=4: -3832.13	F=5: -479.45	F=6: +3525.97	F=7: +8174.44
b ³ F _{5/2}	810.0(24)	-60.00	F=1: -15620.77	F=2: -13992.12	F=3: -11550.93	F=4: -8299.34	F=5: -4240.21	F=6: +622.89	F=7: +6285.67	F=8: +12743.13
z ¹ F _{5/2}	655.4(24)	30.00	F=0: -10305.69	F=1: -9653.40	F=2: -8348.21	F=3: -6388.90	F=4: -3773.62	F=5: -499.93	F=6: +3435.23	F=7: +8035.54
z ³ F _{7/2}	772.0(21)	210.00	F=2: -17475.45	F=3: -15193.91	F=4: -12144.21	F=5: -8319.81	F=6: -3712.54	F=7: +1687.43	F=8: +7891.56	F=9: +14912.92
z ³ F _{5/2}	559.1(21)	30.00	F=1: -6273.92	F=2: -5164.27	F=3: -3496.58	F=4: -1267.00	F=5: +1529.62	F=6: +4899.71		
z ³ G _{5/2}	517.1(27)	180.00	F=1: -9872.44	F=2: -8863.88	F=3: -7345.67	F=4: -5311.40	F=5: -2752.49	F=6: +341.78	F=7: +3984.26	F=8: +8189.95
z ³ D _{5/2}	751.0(27)		F=0: -11827.93	F=1: -11076.95	F=2: -9575.00	F=3: -7322.06	F=4: -4318.14	F=5: -562.24	F=6: +3942.64	F=7: +9199.50
z ³ G _{3/2}	648.2(15)	150.00	F=0: -6978.64	F=1: -6545.75	F=2: -5676.92	F=3: -4366.03	F=4: -2603.88	F=5: -378.23	F=6: +2326.21	F=7: +5527.83
z ³ G _{1/2}	696.1(27)	30.00	F=1: -7815.28	F=2: -6431.61	F=3: -4352.89	F=4: -1575.27	F=5: +1906.40	F=6: +6098.55		
z ³ G _{5/2}	398.4(18)	150.00	F=1: -4401.87	F=2: -3647.89	F=3: -2500.84	F=4: -941.45	F=5: +1056.01	F=6: +3523.68		
z ³ D _{3/2}	705(6)	30.00	F=2: -4739.37	F=3: -2647.27	F=4: +162.20	F=5: +3706.18				
z ³ G _{7/2}	495(3)		F=1: -9522.20	F=2: -8532.89	F=3: -7048.90	F=4: -5070.27	F=5: -2596.97	F=6: +371.00	F=7: +3833.61	F=8: +7790.90
z ³ F _{5/2}	448.8(18)	180.00	F=0: -6972.01	F=1: -6541.59	F=2: -5677.07	F=3: -4371.11	F=4: -2612.68	F=5: -387.10	F=6: +2324.01	F=7: +5542.68
y ¹ D _{5/2}	478.2(24)	30.00	F=0: -7450.82	F=1: -6987.96	F=2: -6059.17	F=3: -4658.33	F=4: -2776.26	F=5: -400.72	F=6: +2483.61	F=7: +5895.08
y ³ G _{5/2}	300(6)		F=2: -6820.22	F=3: -5920.85	F=4: -4721.69	F=5: -3222.74	F=6: -1424.00	F=7: +674.53	F=8: +3072.85	F=9: +5770.96
y ¹ G _{3/2}	334.3(27)		F=1: -6434.70	F=2: -5766.16	F=3: -4763.35	F=4: -3426.27	F=5: -1754.92	F=6: +250.70	F=7: +2590.59	F=8: +5264.75
y ³ D _{5/2}	465(6)		F=1: -5227.65	F=2: -4298.29	F=3: -2904.25	F=4: -1045.53	F=5: +1277.87	F=6: +4065.95		
y ³ F _{7/2}	297(3)	240.00	F=1: -5603.21	F=2: -5043.91	F=3: -4197.83	F=4: -3056.38	F=5: -1608.15	F=6: +161.16	F=7: +2268.69	F=8: +4734.44
y ³ F _{5/2}	508.1(2)		F=0: -8003.36	F=1: -7495.21	F=2: -6478.91	F=3: -4954.46	F=4: -2921.86	F=5: -381.11	F=6: +2667.79	F=7: +6224.84
y ¹ G _{1/2}	441.6(18)		F=1: -8500.61	F=2: -7617.43	F=3: -6292.66	F=4: -4526.30	F=5: -2318.35	F=6: +331.19	F=7: +3422.32	F=8: +6955.04
y ³ G _{3/2}	387(6)		F=0: -6091.00	F=1: -5704.27	F=2: -4930.81	F=3: -3770.62	F=4: -2223.70	F=5: -290.05	F=6: +2030.33	F=7: +4737.44
y ³ D _{3/2}	804.9(27)	-120.00	F=1: -9119.86	F=2: -7475.69	F=3: -5022.30	F=4: -1775.12	F=5: +2245.30	F=6: +7013.23		
(z3946) _{5/2}	603(3)		F=1: -6779.02	F=2: -5573.86	F=3: -3766.12	F=4: -1355.81	F=5: +1657.10	F=6: +5272.57		
z ³ D _{5/2}	1190.8(24)	-60.00	F=2: -8070.04	F=3: -4454.79	F=4: +325.56	F=5: +6236.70				
z ³ D _{3/2}	303.7(21)		F=0: -4783.12	F=1: -4479.43	F=2: -3872.05	F=3: -2960.98	F=4: -1746.22	F=5: -227.77	F=6: +1594.37	F=7: +3720.20
z ³ D _{1/2}	355.2(27)	30.00	F=1: -3980.49	F=2: -3278.56	F=3: -2222.45	F=4: -808.31	F=5: +969.01	F=6: +3115.94		
z ³ D _{5/2}	528(6)		F=2: -3561.50	F=3: -1978.61	F=4: +131.91	F=5: +2770.06				
z ³ D _{3/2}	1673(9)		F=3: -3763.80	F=4: +2927.40						
z ³ F _{5/2}	441(6)		F=1: -8483.28	F=2: -7601.90	F=3: -6279.83	F=4: -4517.07	F=5: -2313.62	F=6: +330.52	F=7: +3413.35	F=8: +6940.87
z ³ G _{1/2}	237(6)	240.00	F=2: -5288.84	F=3: -4617.59	F=4: -3713.87	F=5: -2570.19	F=6: -1177.20	F=7: +476.32	F=8: +2403.45	F=9: +4619.17
z ³ F _{7/2}	531(6)		F=0: -8074.24	F=1: -7561.59	F=2: -6536.29	F=3: -4998.34	F=4: -2947.74	F=5: -384.49	F=6: +2691.41	F=7: +6279.96
z ³ G _{3/2}	330(3)		F=1: -6348.07	F=2: -5688.53	F=3: -4699.22	F=4: -3380.14	F=5: -1731.29	F=6: +247.33	F=7: +2555.72	F=8: +5193.88
z ³ F _{5/2}	660(24)		F=1: -7419.82	F=2: -6100.74	F=3: -4122.12	F=4: -1483.96	F=5: +1813.73	F=6: +5770.98		
z ³ F _{3/2}	1019(9)		F=2: -6880.27	F=3: -3822.38	F=4: +254.82	F=5: +351.32				
z ³ G _{5/2}	537(27)		F=0: -8451.92	F=1: -7915.29	F=2: -6842.03	F=3: -5232.14	F=4: -3085.62	F=5: -402.47	F=6: +2817.31	F=7: +6573.72
z ³ G _{3/2}	929(9)		F=1: -10455.30	F=2: -8596.58	F=3: -5808.50	F=4: -2091.06	F=5: -2555.74	F=6: +8131.90		
y ¹ D _{3/2}	681(6)	60.00	F=1: -7623.82	F=2: -6279.90	F=3: -4257.60	F=4: -1549.19	F=5: +1855.60	F=6: +4569.64		
y ¹ F _{5/2}	983(6)	-150.00	F=1: -18997.66	F=2: -17009.59	F=3: -14031.95	F=4: -10070.10	F=5: -5131.18	F=6: +775.88	F=7: +7640.37	F=8: +15449.79
y ¹ D _{1/2}	711(6)		F=0: -11190.53	F=1: -10480.02	F=2: -9059.00	F=3: -6927.47	F=4: -4085.43	F=5: -532.88	F=6: +3730.18	F=7: +8703.75
y ³ F _{7/2}	711(9)	-150.00	F=0: -11270.89	F=1: -10545.07	F=2: -9096.50	F=3: -6931.30	F=4: -4058.65	F=5: -490.79	F=6: +3756.96	F=7: +8666.25
(43848) _{5/2}	779(3)		F=0: -12276.50	F=1: -11497.03	F=2: -9938.11	F=3: -7599.73	F=4: -4481.90	F=5: -584.60	F=6: +4092.16	F=7: +9548.39
y ¹ G _{1/2}	982.1(27)	-150.00	F=2: -22405.28	F=3: -19434.37	F=4: -15478.62	F=5: -10542.69	F=6: -4632.44	F=7: +2225.13	F=8: +10081.83	F=9: +18868.31
(44202) _{5/2}	660(24)		F=1: -7419.82	F=2: -6100.74	F=3: -4122.12	F=4: -1483.96	F=5: +1813.73	F=6: +5770.98		
y ¹ G _{3/2}	546(12)		F=0: -8593.51	F=1: -8047.90	F=2: -6956.65	F=3: -5319.80	F=4: -3137.32	F=5: -409.21	F=6: +2864.51	F=7: +6683.85
y ¹ F _{3/2}	435(9)	180.00	F=2: -2837.80	F=3: -1662.27	F=4: +25.10	F=5: +327.18				

Measured A, B constants; sources in the data file.